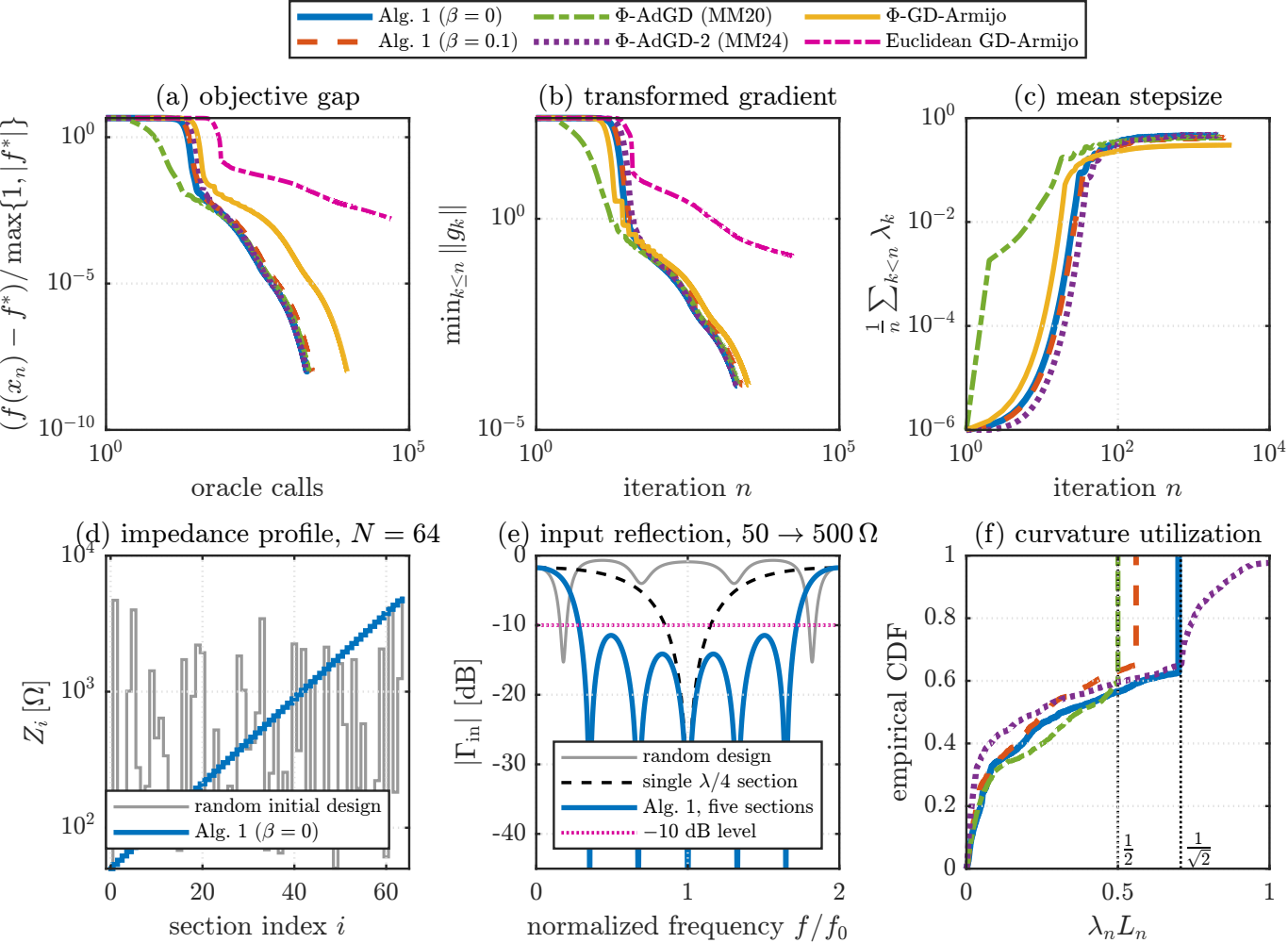

Alg. 1 ($\beta = 0$)
Alg. 1 ($\beta = 0.1$)
Φ-AdGD (MM20)
Φ-AdGD-2 (MM24)
Φ-GD-Armijo
Euclidean GD-Armijo
(a) objective gap
$(f(x_n) - f^*)/\max\{1, |f^*|\}$
oracle calls
(b) transformed gradient
$\min_{k \le n} \|g_k\|$
iteration n
(c) mean stepsize
$\frac{1}{n}\sum_{k<n} \lambda_k$
iteration n
(d) impedance profile, $N = 64$
Z_i [Ω]
random initial design
Alg. 1 ($\beta = 0$)
section index i
(e) input reflection, 50 → 500 Ω
$|\Gamma_{\text{in}}|$ [dB]
random design
single $\lambda/4$ section
Alg. 1, five sections
−10 dB level
normalized frequency f/f_0
(f) curvature utilization
empirical CDF
$\frac{1}{2}$
$\frac{1}{\sqrt{2}}$
$\lambda_n L_n$